Muhammad Ehab Sikandar

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Linkedin

Education

University of Central Punjab
Degree in Bachelor's of Science in Computer Science — Oct 2020 - Aug 2024

Experience

CodeBricks Global | AI Research Assistant — May 2025 - Present

- Designed and implemented end-to-end analytical pipelines in Python and R to process and analyze large-scale consumer survey datasets.
- Applied PLSR, PCA, correlation analysis, and Random Forest models to identify key drivers of consumer liking and purchase intent.
- Built driver analysis frameworks to quantify the impact of product attributes, enabling data-driven product optimization decisions.
- Developed ideal profiling models using regression techniques to estimate optimal sensory attribute levels and identify gaps between current and ideal product characteristics.
- Performed consumer segmentation and preference mapping using PCA and clustering, uncovering distinct audience segments and product positioning insights.
- Automated survey data preprocessing workflows, reducing manual effort and improving consistency and reliability of analytics outputs.
- Conducted statistical analysis and hypothesis testing using Python, R, and SPSS to validate insights across multiple methodologies.
- Created data visualizations and analytical reports using Power BI and Excel to communicate findings to non-technical stakeholders.

VentureDive | Data Science Intern — Jan 2025 - Mar 2025

- Contributed to the development of an LLM-powered financial data analysis system for querying structured CSV datasets.
- Implemented SQL-like querying logic using Pandas, enabling filtering, aggregation, and trend analysis on financial data.
- Designed a dataframe-based query pipeline for extracting insights without reliance on vector search.
- Supported data cleaning, preprocessing, and transformation, ensuring high-quality structured datasets for analysis.
- Improved interpretability of model outputs by incorporating explainable reasoning in query responses.

Skills

Programming Languages:	Python, R
Libraries/Frameworks:	Pandas, NumPy, Scikit-learn, statsmodels, PyTorch, Hugging Face Transformers
Tools / Platforms:	VS Code, Power BI, Excel, SPSS, Jupyer Notebook
Databases:	SQL

Projects / Open-Source

Consumer Preference Driver Analysis — *Python, R, PLSR, Random Forest, Excel*

- Built machine learning and statistical models (PLSR, Random Forest) to identify key product attributes influencing consumer liking and purchase intent.
- Developed pipelines to preprocess survey data and evaluate attribute importance across multiple techniques.
- Delivered actionable insights enabling product teams to prioritize high-impact features.
- Validated and cross-verified model outputs across multiple statistical techniques, increasing confidence in identified drivers and supporting reliable decision-making.

Ideal Profiling Analysis — *Python, R, Regression Analysis*

- Developed regression-based models to estimate optimal sensory attribute levels that maximize consumer liking.
- Identified gaps between current and ideal product profiles to guide product reformulation strategies.

Consumer Segmentation & Preference Mapping *Python, R, PCA, Clustering, Power BI*

- Applied PCA and clustering techniques to identify consumer segments based on behavioral and preference patterns.
- Created visual preference maps to analyze product positioning within competitive categories.

Survey Analytics Workflow Automation *Python, Excel*

- Designed automated data pipelines for cleaning, validation, and structuring of survey datasets.
- Reduced manual preprocessing effort and improved turnaround time for analytics delivery.

LibraBot Offline LLM-Based Virtual Assistant *Python, PyTorch, Transformers, Flask, OCR, NVIDIA Omniverse*

- Developed an offline AI-powered virtual assistant using transformer-based LLMs for conversational query handling.
- Integrated OCR for text extraction and NVIDIA Audio2Face for avatar-based interaction.
- Deployed a local Flask application for secure, real-time query processing without cloud dependency.